

MID

Subject: computer networks

Paper: MID - Set A | Duration: 2 Hours | Max Marks: 37 | Date: 14-03-2026

Total Questions: 11 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQS

1. Which layer of the OSI model is responsible for logical addressing (IP addresses)? [2 Marks]
 - a) Physical Layer
 - b) Data Link Layer
 - c) Network Layer
 - d) Transport Layer

2. Which of the following protocols is considered connection-oriented? [2 Marks]
 - a) UDP
 - b) HTTP
 - c) TCP
 - d) ICMP

3. What is the primary function of a modem in a computer network? [2 Marks]
 - a) To route data packets between networks.
 - b) To modulate and demodulate signals for transmission over analog lines.
 - c) To filter network traffic and improve security.
 - d) To connect multiple devices within a local area network.

4. A MAC address uniquely identifies a device on a network segment. What is its typical length? [2 Marks]
 - a) 16 bits
 - b) 32 bits
 - c) 48 bits
 - d) 128 bits

5. In the context of computer networks, what does LAN stand for? [2 Marks]
 - a) Local Area Network
 - b) Large Access Node
 - c) Link Area Node
 - d) Low-latency Access Network

6. Which of the following transmission media is least susceptible to electromagnetic interference (EMI)? [3 Marks]
 - a) Coaxial Cable
 - b) Unshielded Twisted Pair (UTP)
 - c) Fiber Optic Cable

d) Shielded Twisted Pair (STP)

SECTION A: SHORT ANSWERS

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| 7. Define the primary purpose of the Transport Layer in the OSI model. | [3 Marks] |
| 8. Explain the fundamental concept of 'packet switching' in computer networks. | [3 Marks] |
| 9. Differentiate between a hub and a switch based on how they handle network traffic. | [5 Marks] |
| 10. Briefly describe two common network topologies and their main characteristics. | [6 Marks] |
| 11. Discuss the role of DNS (Domain Name System) in network communication. | [7 Marks] |

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